

0.1 All-electron Kohn–Sham accuracy

We next assess the accuracy of SPARC-atomSFE for all-electron Kohn–Sham DFT calculations through comparisons with literature codes. In particular, we consider four exchange–correlation functionals: LDA, PBE, rSCAN, and HF. For LDA, PBE, and rSCAN, results are compared for ten atoms ranging from light to heavy—H, Be, C, Ne, Na, Si, Fe, Kr, Gd, and U—against the FeAtom code for LDA and the atomPAW code for PBE and rSCAN. For HF, comparisons are carried out for closed-shell neutral atoms against Lehtola *et al.*, and for charged species—anions H^- , Li^- , F^- , Na^- , Cl^- and cations Li^+ , B^+ , Na^+ , Al^+ —against Lehtola *et al.* The distribution of the differences in the results is shown as violin plots in Figure 1. We observe that total energies and occupied eigenvalues agree to 10^{-6} Ha or better in most cases; the somewhat larger errors observed for PBE and rSCAN can likely be attributed to the accuracy of the atomPAW reference results (as we were unable to converge them further) rather than to SPARC-atomSFE itself, with rSCAN showing slightly worse agreement than PBE. Notably, for LDA, the total energy and eigenvalues of SPARC-atomSFE and FeAtom agree to 10^{-8} Ha and 10^{-9} Ha, respectively, even for the heaviest atom considered, U ($Z = 92$). These results demonstrate the accuracy of SPARC-atomSFE for atomic structure calculations based on all-electron Kohn–Sham DFT.

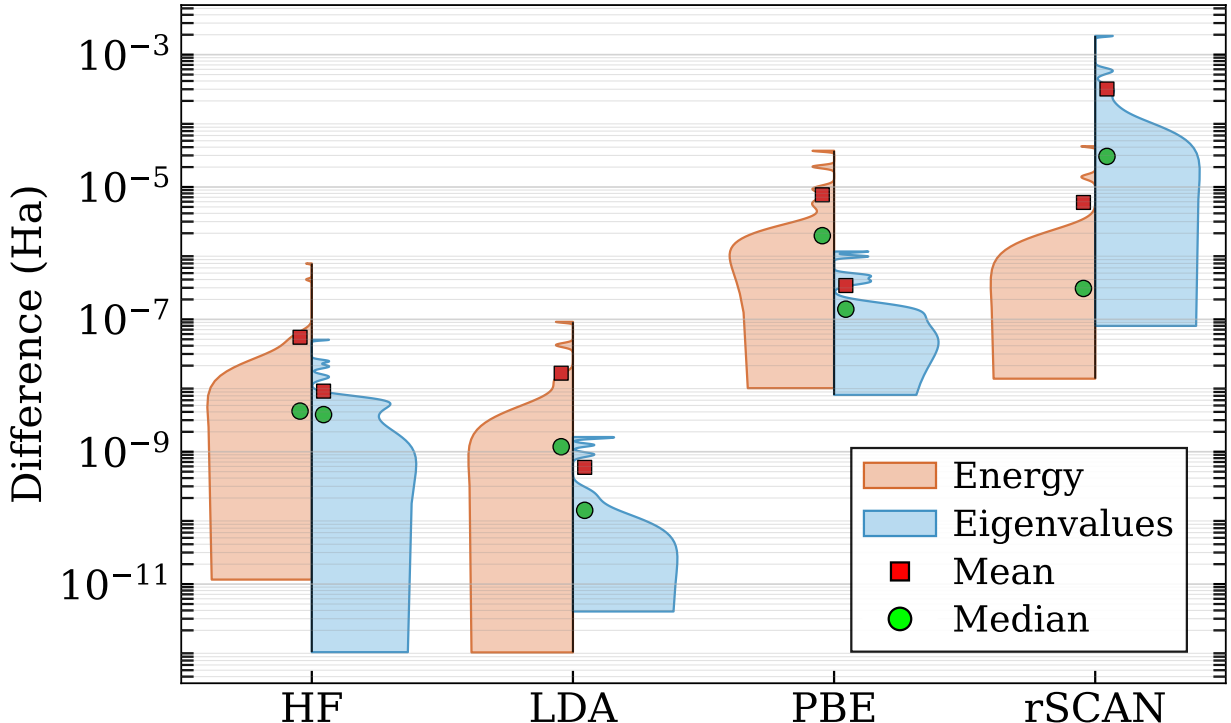


Figure 1: Comparison of SPARC-atomSFE all-electron Kohn–Sham DFT results with literature for various exchange–correlation functionals. The energy and eigenvalue errors denote the absolute differences in the corresponding quantities.

As supplementary material to the accuracy discussion above, we tabulate the full numerical data underlying Figure 1.

Table 1: Comparison of total energies and eigenvalues for the choice of LDA-SVWN exchange-correlation functional.

Z		$-E_{\text{tot}}$ (Ha)			Occupied-eigenvalue Difference (Ha)	
		atomSFE	FeAtom	Difference	Max $ \Delta\varepsilon $	Mean $ \Delta\varepsilon $
H	1	0.44567051824	0.44567051824	9.29×10^{-13}	3.86×10^{-12}	3.86×10^{-12}
Be	4	14.4472094741	14.4472094740	8.31×10^{-11}	3.10×10^{-11}	1.90×10^{-11}
C	6	37.4257485361	37.4257485360	1.98×10^{-10}	9.68×10^{-11}	4.20×10^{-11}
Ne	10	128.233481270	128.233481269	7.44×10^{-10}	3.64×10^{-10}	2.17×10^{-10}
Na	11	161.440060310	161.440060309	9.04×10^{-10}	6.99×10^{-11}	4.39×10^{-11}
Si	14	288.198396604	288.198396603	1.48×10^{-09}	1.37×10^{-11}	8.03×10^{-12}
Fe	26	1261.09305585	1261.09305584	5.67×10^{-09}	2.04×10^{-09}	1.26×10^{-09}
Kr	36	2750.14794043	2750.14794042	1.22×10^{-08}	1.50×10^{-09}	9.08×10^{-10}
Gd	64	10816.6538772	10816.6538772	4.09×10^{-08}	2.37×10^{-09}	1.67×10^{-09}
U	92	25658.4178889	25658.4178888	9.18×10^{-08}	2.67×10^{-09}	1.63×10^{-09}

Table 2: Comparison of total energies and eigenvalues for the choice of PBE exchange-correlation functional.

Z		$-E_{\text{tot}}$ (Ha)			Occupied-eigenvalue Difference (Ha)	
		atomSFE	atomPAW	Difference	Max $ \Delta\varepsilon $	Mean $ \Delta\varepsilon $
H	1	0.458928588	0.458928597	9.13×10^{-09}	7.22×10^{-09}	7.22×10^{-09}
Be	4	14.62994793	14.62994772	2.14×10^{-07}	9.80×10^{-08}	5.27×10^{-08}
C	6	37.74820924	37.74820878	4.55×10^{-07}	4.04×10^{-08}	2.95×10^{-08}
Ne	10	128.8664291	128.8664277	1.38×10^{-06}	1.34×10^{-07}	9.12×10^{-08}
Na	11	162.1645960	162.1645945	1.55×10^{-06}	2.51×10^{-07}	1.32×10^{-07}
Si	14	289.2027550	289.2027529	2.13×10^{-06}	5.84×10^{-07}	1.53×10^{-07}
Fe	26	1263.295438	1263.295432	5.63×10^{-06}	2.21×10^{-06}	3.70×10^{-07}
Kr	36	2753.416118	2753.416109	9.33×10^{-06}	1.37×10^{-06}	4.58×10^{-07}
Gd	64	10823.19862	10823.19860	2.03×10^{-05}	8.75×10^{-06}	1.06×10^{-06}
U	92	25668.41171	25668.41167	3.53×10^{-05}	1.05×10^{-05}	8.92×10^{-07}

Table 3: Comparison of total energies and eigenvalues for the choice of rSCAN exchange-correlation functional.

Z		$-E_{\text{tot}}$ (Ha)			Occupied-eigenvalue Difference (Ha)	
		atomSFE	atomPAW	Difference	Max $ \Delta\varepsilon $	Mean $ \Delta\varepsilon $
H	1	0.452860365	0.452860352	1.27×10^{-08}	7.97×10^{-08}	7.97×10^{-08}
Be	4	14.65119802	14.65119794	8.55×10^{-08}	8.66×10^{-06}	4.68×10^{-06}
C	6	37.76635704	37.76635699	5.21×10^{-08}	1.34×10^{-05}	5.75×10^{-06}
Ne	10	128.9723925	128.9723925	4.62×10^{-08}	5.24×10^{-05}	2.29×10^{-05}
Na	11	162.2862159	162.2862160	6.15×10^{-08}	6.51×10^{-05}	2.19×10^{-05}
Si	14	289.3767993	289.3767988	5.01×10^{-07}	1.22×10^{-04}	3.51×10^{-05}
Fe	26	1263.668535	1263.668536	1.04×10^{-06}	6.72×10^{-04}	1.53×10^{-04}
Kr	36	2754.194549	2754.194547	1.16×10^{-06}	1.39×10^{-03}	2.97×10^{-04}
Gd	64	10824.64115	10824.64117	1.42×10^{-05}	3.08×10^{-03}	5.69×10^{-04}
U	92	25670.94593	25670.94589	4.13×10^{-05}	1.42×10^{-02}	1.92×10^{-03}

Table 4: Hartree–Fock closed-subshell benchmark for 17 atoms: SPARC-atomSFE vs. Lehtola *et al.* Entries are total energy E_{tot} , HOMO eigenvalue $\varepsilon_{\text{HOMO}}$, and exchange energy E_x (Ha).

Z	$-E_{\text{tot}}$ (Ha)			$-\varepsilon_{\text{HOMO}}$ (Ha)			$-E_x$ (Ha)		
	atomSFE	Lehtola	Difference	atomSFE	Lehtola	Difference	atomSFE	Lehtola	Difference
He 2	2.861679995600	2.861679995612	1.17×10^{-11}	0.91795556285	0.91795556286	2.57×10^{-12}	1.025768869910	1.025768869900	1.04×10^{-11}
Be 4	14.57302316828	14.57302316832	3.60×10^{-11}	0.30926955153	0.30926955157	4.15×10^{-11}	2.666913669350	2.666913669390	3.96×10^{-11}
Ne 10	128.5470981088	128.5470981094	5.79×10^{-10}	0.85040965035	0.85040965035	9.37×10^{-13}	12.10835073105	12.10835073117	1.15×10^{-10}
Mg 12	199.6146364237	199.6146364245	7.70×10^{-10}	0.25305258176	0.25305258201	2.51×10^{-10}	15.99429168655	15.99429168694	3.91×10^{-10}
Ar 18	526.8175128010	526.8175128027	1.76×10^{-9}	0.59101740571	0.59101740935	3.64×10^{-9}	30.18494199109	30.18494198731	3.78×10^{-9}
Ca 20	676.7581859224	676.7581859245	2.12×10^{-9}	0.19552969154	0.19552969247	9.30×10^{-10}	35.21120850330	35.21120850517	1.87×10^{-9}
Zn 30	1777.848116185	1777.848116191	6.10×10^{-9}	0.29250714776	0.29250714640	1.36×10^{-9}	69.64119795154	69.64119795404	2.50×10^{-9}
Kr 36	2752.054977337	2752.054977346	8.62×10^{-9}	0.52418666755	0.52418668103	1.35×10^{-8}	93.85599601220	93.85599600354	8.66×10^{-9}
Sr 38	3131.545686429	3131.545686439	9.30×10^{-9}	0.17845644076	0.17845644685	6.09×10^{-9}	101.9500832606	101.9500832705	9.84×10^{-9}
Pd 46	4937.921024056	4937.921024070	1.44×10^{-8}	0.33600200815	0.33600201367	5.52×10^{-9}	139.1429137365	139.1429137267	9.78×10^{-9}
Cd 48	5465.133142514	5465.133142530	1.57×10^{-8}	0.26485590698	0.26485591255	5.58×10^{-9}	148.9142513282	148.9142513291	9.70×10^{-10}
Xe 54	7232.138363852	7232.138363872	1.95×10^{-8}	0.45729006132	0.45729005785	3.47×10^{-9}	179.0971093189	179.0971093669	4.80×10^{-8}
Ba 56	7883.543827310	7883.543827330	2.09×10^{-8}	0.15752769303	0.15752769864	5.61×10^{-9}	189.1002498282	189.1002498415	1.33×10^{-8}
Yb 70	13391.45619308	13391.45619312	3.60×10^{-8}	0.18246254162	0.18246252223	1.94×10^{-8}	276.2138392544	276.2138392265	2.79×10^{-8}
Hg 80	18408.99149490	18408.99149494	4.50×10^{-8}	0.26104596502	0.26104601419	4.92×10^{-8}	345.3035177000	345.3035163829	1.32×10^{-6}
Rn 86	21866.77224082	21866.77224087	5.30×10^{-8}	0.42800677179	0.42800676968	2.12×10^{-9}	387.5037738100	387.5037737909	1.91×10^{-8}
Ra 88	23094.30366637	23094.30366642	5.50×10^{-8}	0.14877117348	0.14877119673	2.33×10^{-8}	401.4148473300	401.4148474301	1.00×10^{-7}

Table 5: Hartree–Fock charged benchmark: SPARC-atomSFE vs. Lehtola *et al.* Values are reported as $-E_{\text{tot}}$ (Ha). Since the reference table is reported to 9 digits after the decimal point, the absolute total-energy errors are computed from these rounded values and reported with conservative precision.

	Z	N_e	atomSFE	Lehtola	Difference
H ⁻	1	2	0.487929734	0.487929734	$< 1 \times 10^{-9}$
Li ⁺	3	2	7.236415201	7.236415201	$< 1 \times 10^{-9}$
Li ⁻	3	4	7.428231641	7.428232061	4×10^{-7}
B ⁺	5	4	24.237575184	24.237575184	$< 1 \times 10^{-9}$
F ⁻	9	10	99.459453912	99.459453913	1×10^{-9}
Na ⁺	11	10	161.676962614	161.676962614	$< 1 \times 10^{-9}$
Na ⁻	11	12	161.855125277	161.855125996	7×10^{-7}
Al ⁺	13	12	241.674670464	241.674670465	1×10^{-9}
Cl ⁻	17	18	459.576925266	459.576925268	2×10^{-9}